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ORIGINAL ARTICLE

Assessment of springwater quality for drinking and irrigation in Haut Plateaux Division, West Region of Cameroon

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Abstract

Springs serve as essential water sources for drinking and small-scale agriculture for most rural residents in the Haut Plateau division. Unfortunately, human activities are increasingly compromising these springwater quality throughout the division. This research seeks to assess springwater quality in the Haut Plateau Division and determine whether these water sources remain suitable for consumption and agricultural irrigation practices using physicochemical parameters, water quality index, irrigation indices and irrigation water quality index. A total of 37 springwater samples were collected from the study area in March 2022 and analysed to determine their hydrochemical characteristics and to evaluate the hydrochemical evolution processes of springwater in the area. The findings revealed that pH levels in 70.3% of the springwater samples were slightly acidic (ranging from 4.2 to 7.7), while NO_3^- concentrations exceeded standard recommended limits in 16.22% of the samples collected. However, water quality index classification indicated that 79.4% of samples were rated as excellent, while 20.6% as good for drinking purposes. Hydrochemical facies showed the mixed Ca–Mg–Cl as the predominant water type, thus revealing rock–water interaction and reverse ion exchange as dominant processes controlling the springwater quality. Regarding spring water quality for irrigation, several indices were evaluated, including sodium adsorption ratio, sodium percentage, Kelly's ratio, residual sodium carbonate, magnesium adsorption ratio, permeability index and corrosivity ratio. Results showed that 5.7%, 91.1%, 89.2%, and 45.9% were unsuitable based on Kelly's ratio, magnesium adsorption ratio, Corrosivity ratio, and permeability index. However, the irrigation water quality index revealed that 86.5% of the springwater samples in the study area were good for irrigation use. To improve water quality in problem areas, two solutions are suggested: using limestone filters to fix low pH and installing biological systems to reduce high nitrate levels.

Keywords Haut Plateau division · Irrigation · Physicochemical parameters · Springwater · Water quality index

Introduction

Clean, sufficient water is essential for both human civilization and environmental health. As global population grows and cities expand, especially in developing nations, the demand for water continues to surge (Ahada & Suthar, 2018; Nong et al., 2020). It is reported that over five billion people rely on groundwater and surface water systems for their daily needs, including drinking water, residential use, farming, and industrial processes (Fallatah & Khattab, 2023). However, both developing and developed nations struggle with water quality challenges. In developing countries, the primary struggle lies in establishing and maintaining basic water supply and sanitation infrastructure while simultaneously protecting water quality (Mbinkong et al., 2024a, 2024b; Nzouemou et al., 2025). Meanwhile, developed countries work to combat issues like nutrient pollution and

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algal blooms in their water bodies, while also meeting the demands of growing populations (Fano et al., 1986; Hwang et al., 2024).

Water covers approximately 71% of Earth's surface and is among the planet's most vital natural resources. It exists in various forms, including oceans, rivers, lakes, glaciers, and groundwater (Mishra et al., 2023). Despite this abundance, freshwater represents just about 2.5% of all water, yet must sustain human consumption, farming, industrial activities, and ecosystem functions. Freshwater exists in two main forms: groundwater and surface water, with surface water being readily accessible in rivers, streams, lakes, ponds, estuaries and springs. Some of these surface water sources serve as the main source of water for different needs in semi-urban and rural areas of developing nations where proper water infrastructure is absent. However, water quality in surface water bodies has become a global concern, more so than groundwater sources. The condition of surface waters is affected by both natural factors (such as climate change, interactions between water and rock formations, and geological conditions) and human activities (including agricultural practices, urban waste disposal and industrial activities). This vulnerability affects both human well-being and aquatic ecosystems (Egbueri, 2020; Gopinath et al., 2013; Makhlouf et al., 2025). The economic implications are substantial, affecting sectors such as agriculture, fisheries, tourism, and public health. Due to the widespread impact of water quality issues on daily life, ecosystem health, and their prevalence across multiple countries, the implementation of consistent water quality monitoring programs has become crucial for protecting these essential freshwater resources (Mohammadi et al., 2022). These monitoring efforts are often complemented by strict regulatory frameworks, watershed management practices, and public awareness campaigns to ensure sustainable use and protection of surface water resources for future generations (Dehghani et al., 2022).

Managing water resources involves handling and scrutinising extensive datasets related to water quality, which can be challenging to assess and consolidate. Scientists have often used direct comparison of water parameters with standard recommended limits to determine its suitability for various purposes (Alghamdi et al., 2023; Kumar et al., 2022; Lanjwani et al., 2022; Ustaoğlu et al., 2020; Zemunac et al., 2021). Under this approach, water is considered unsuitable for its intended use if any parameter exceeds its prescribed limit. However, due to discrepancies in the importance of water parameters, the Water Quality Index (WQI) technique was born to evaluate overall water quality for various purposes by converting multiple parameters into one numerical value (Brown et al., 1970). This index weights different water quality measurements based on their importance, creating a single score that is easily understood by the public, water managers, and policymakers alike (Abdessamed et al., 2023; Howladar et al., 2018; Sultana et al., 2025). Moreover, irrigation indices such as SAR, Na%, RSC, KI, MH, PI and CR has been adopted for use in assessing water quality for irrigation purposes (Al-Aizari et al., 2024; Mbinkong al., 2024a, b; Rawat et al., 2018; Xu et al., 2019).

In the Haut Plateaux Division, West Region of Cameroon, residents in rural areas heavily depend on traditional spring-water resources for both drinking and small-scale agriculture due to the lack of a sustainable water supply network. Springs are natural water sources found throughout various landscapes worldwide. They occur through a natural process where groundwater emerges at the earth's surface by flowing through geological features such as fractures, faults, or rock interfaces (Marghade et al., 2021). Springs can be classified as perennial or short-lived. Perennial springs flow continuously, drawing water from below the groundwater table, while short-lived springs are active temporarily after the rainy seasons when groundwater levels rise. The discharge rate, volume, and water quality of springs are determined by the hydrogeological and geochemical characteristics of the spring-shed—the area that drains into the spring (Aouiti et al., 2021; von Fumetti et al., 2017). While springwater is generally considered pure and cost-effective due to natural soil filtration, its actual quality depends on specific physical, chemical and biological measurements, and human activities can progressively alter these properties, causing widespread negative impacts throughout the entire ecosystem. In the study area, there is growing concern, as over 80% of the springs, crucial for drinking and agriculture, are underdeveloped, raising questions about their safety and usability. Therefore, assessing the springwater quality is essential, as it has been a consistent focus of scientific research over the years (Abduljabar et al., 2020; Dumaru et al., 2021; Taloor et al., 2020). Such studies have established baseline data in previously unstudied areas, providing insights into interventions needed for spring-shed conservation. Previous research focused on groundwater quality assessment has been conducted in other parts of Cameroon's West region (Joelle et al., 2021; Mba et al., 2019; Mfonka et al., 2021; Mufur et al., 2022; Sanou et al., 2015; Temgoua, 2011; Temgoua et al., 2009). In the study area, the water quality of a well, 2 boreholes and a stream has been determined by Akaho et al. (2022) in the Baham community, based on directly comparing ion concentrations against standards set by the World Health Organization (WHO, 2011). The study reported acidic water and very high concentrations of sulphate and lead, which rendered the water unsuitable for drinking purposes. However, springwater hydrochemistry and its suitability for drinking and irrigation purposes were not examined. Thus, the region continues to face springwater quality degradation in specific areas due to poor household and animal waste management, excessive fertiliser use, and uncontrolled irrigation practices. Therefore, the current

study aims to evaluate springwater quality in the Haut Plateaux Division (Bamenjou, Batie and Baham subdivisions) and determine its suitability for drinking and irrigation uses using hydrochemical data collected in March 2022.

Materials and methods

The study area

The study area is located in the Hauts Plateaux Division of the West Region of Cameroon between 10°16′–10°26′ East longitude and 5°14′–5°27′ North latitude (Fig. 1) with an altitude that ranges from 1000 m in the Noun plain and 2740 m at Mount Meletan in the Bamboutos. The Hauts Plateaux Division covers about 9,000 km^2 with a population density of 128.5 inhabitants per km^2. The region features rugged terrain characterised by plateaus and prominent massifs (including Mount Léfo at 2,550 m, Bamboutos at 2,700 m, and Mbam at 2,335 m), and is covered by periforest savannah with trees and shrubs interspersed with forest massifs and small forest galleries of hydrophilic and hydrophobic plants (Sokeng et al., 2016). The soils are ferrallitic, hydromorphic and poorly evolved. These soils are very fertile and support year-round crop cultivation, while its extensive hydrography, marked by numerous waterfalls and rectilinear drains, maintains constant soil humidity and sustains year-round vegetation under a humid tropical highland climate characterised by two distinct seasons: a dry period from mid-November to mid-March, and a rainy period from mid-March to mid-November. The average annual temperature oscillates around 21 °C, while the average annual rainfall is 1,800 mm. The geology of the study area is made up of a granitic-gneissic bedrock overlain by volcanic deposits. This basement rock primarily comprises heterogeneous granulites (high-grade gneiss), extensive amphibolites, and granitoids, including large plutonic formations like the Bangangté syenite, the charnokites of Bandja and the granite of Batié (Nono et al., 2009). These features form part of the Cameroon Volcanic Line (CVL).

Sampling and analysis

Water samples were collected from 37 different locations in Baham, Bamendjou, and Batie in the Haut Plateaux Division of the West Region of Cameroon, in March 2022. These samples were collected in pre-cleaned polyethylene containers with a capacity of 1.5 L from springs used for drinking and small-scale agriculture. The containers were sealed tightly after being completely filled and then stored at 4 °C in a thermos-insulated flask until laboratory analysis. In the field, various physicochemical parameters such as pH, electrical conductivity (EC) and total dissolved solids (TDS) were measured using a multiparameter probe (Orion Star A225). The coordinates of sampling points were recorded using the Global Positioning System (GPS). Water quality

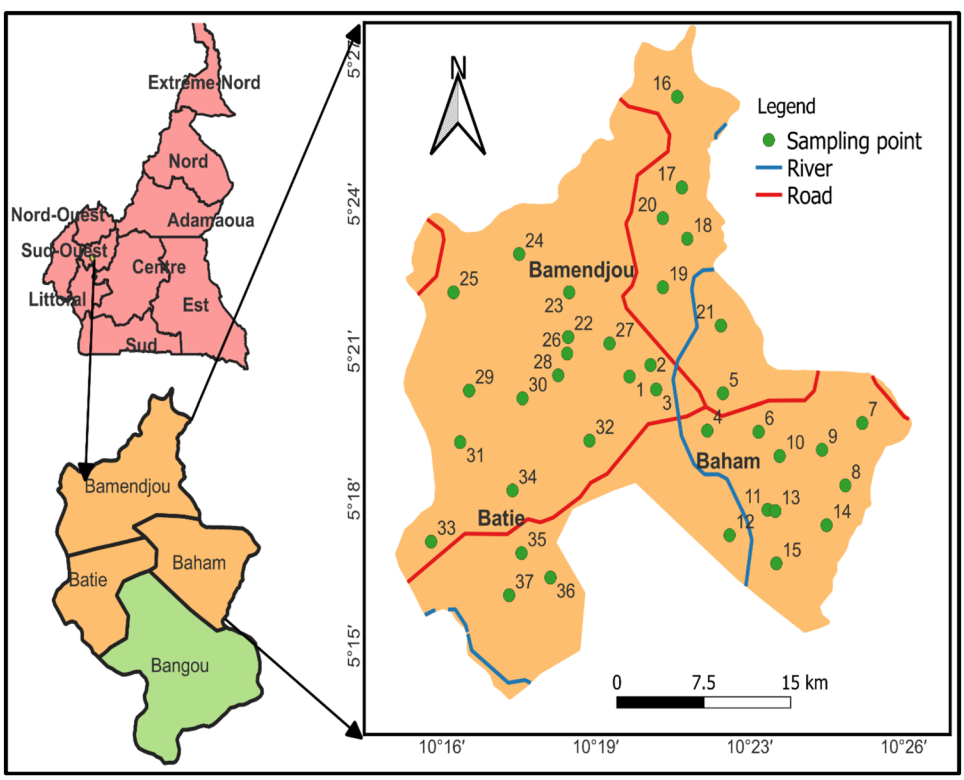

Fig. 1 Location map of the study area showing springwater samples

analyses were conducted at the research unit of soil and environmental chemistry at the University of Dschang. Standard guidelines were followed for laboratory analysis (APHA, 1995), where anions (HCO_3^-, Cl^-, SO_4^{2-}, and NO_3^-) and cations (Ca^{2+}, Mg^{2+}, K^+, and Na^+) were analysed using the Hanna HI83099 multiparameter photometer. This instrument employs photometric techniques to accurately measure the concentration of these ions by passing light through the sample and detecting changes in absorbance, which correlate to the ionic concentrations. Calibration was performed prior to analysis to ensure precision. Total hardness (TH) was calculated from Eq. 1. To ensure analytical accuracy and reliability, charge balance error (CBE) calculations were performed using Eq. (2), and only samples showing CBE values within ±10% were considered for the study.

$$\mathrm{TH}\left(\mathrm{CaCO_3}\right) = 2.497\mathrm{xCa} + 4.118\mathrm{xMg} \tag{1}$$

$$\mathrm{CBE} = \frac{\sum \mathrm{Cations} - \mathrm{Anions}}{\sum \mathrm{Cations} + \mathrm{Anions}} \mathrm{x} \frac{100}{1} \tag{2}$$

Hydrochemical characterisation

The Piper trilinear diagram was used to display and interpret the hydrochemical data, which allows for clear visualisation of water types and chemical relationships in groundwater samples. In addition, Gibbs diagram (Gibbs, 1970) was used to assess the primary controls on groundwater chemistry, specifically examining whether the spring water composition was influenced predominantly by evaporation processes, precipitation inputs, or rock-water interactions. This analysis was performed separately for both cations and anions (measured in milliequivalents per liter) using Eqs. 3 and 4. Furthermore, the possibility of ion exchange processes was evaluated using the chloroalkaline indices (CAI) developed by Schoeller (Schoeller, 1934).

$$\text{Gibbs ratio I} = \frac{\mathrm{Na^+ + K^+}}{\mathrm{Na^+ + K^+ + Ca^{2+}}} \tag{3}$$

$$\text{Gibbs ratio II} = \frac{\mathrm{Cl^-}}{\mathrm{Cl^- + HCO_3^-}} \tag{4}$$

Springwater quality for drinking purposes

The assessment of spring water's potability involved two main approaches: first, individual physiochemical parameters were compared against WHO drinking water standards (WHO, 2011), and second, the weighted arithmetic water quality index (WAWQI) was calculated to evaluate the combined effect of all parameters on overall water quality. This weighted approach assigns unequal importance to each parameter in determining the composite water quality score. The WAWQI calculation consists of three sequential phases to evaluate springwater suitability, based on recommendations. In the first step, weights (w_i) are assigned to the 12 water quality parameters according to their relative importance for drinking water quality (Table 1). The second phase involved calculating the relative weight using Eq. 5. The third step is concerned with evaluating three components: the quality rating (q_i) using Eq. 6, the sub-index (SI) using Eq. 7, and the water quality index using Eq. 8.

$$W_i = \frac{w_i}{\sum_{i=1}^{n} w_i} \tag{5}$$

$$\mathrm{q_i} = \frac{\mathrm{V_i}}{\mathrm{S_i}} \mathrm{x} \frac{100}{1} \tag{6}$$

Table 1 Physiochemical parameters, standard limits, weights and relative weights

Parameter	Unit	Minimum	Maximum	Mean	Standard limit (WHO, 2011)	Weight (w_i)	Relative weight (W_i)
pH	–	4.2	7.7	5.93	6.5–8.5	4	0.098
EC	$\mu S/cm$	38.39	254.32	132.68	1000	5	0.122
TDS	mg/L	22	151	78.92	500	4	0.098
TH	mg/L	11.30	111.24	50.09	500	3	0.073
Na^+	mg/L	0.81	13.40	5.48	200	2	0.049
K+	mg/L	0.13	8.95	3.34	12	2	0.049
Ca^{2+}	mg/L	0.62	20.85	7.80	75	3	0.073
Mg^{2+}	mg/L	0.74	15.41	7.44	50	3	0.073
Cl^-	mg/L	6.13	44.95	20.62	250	3	0.073
HCO_3^-	mg/L	3.54	79.15	29.54	120	3	0.073
SO_4^{2-}	mg/L	0.25	29.22	4.71	250	4	0.098
NO_3^-	mg/L	0.02	56.65	11.71	45	5	0.122

$$SI = q_i x W_i \tag{7}$$

$$\text{IWQI} = \frac{1}{n} x \sum_{i=1}^{n} \text{SI} \tag{8}$$

where W_i denotes the relative weight, q_i represents the quality rating for each parameter i, V_i is the measured/observed value of parameter i, and S_i refers to the standard water quality value. The WAWQI categorised the springwater quality into five classifications: excellent (0–25), good (26–50), poor (51–75), very poor (76–100), and unfit or unsuitable (> 100) (Ameen, 2019).

Springwater quality for irrigation purposes

Irrigation water quality is determined by the amount of dissolved substances present in the water. Optimal crop yields can be achieved through the combination of high-quality irrigation water, proper water management practices, and favorable soil conditions. In this study, the quality of irrigation water was evaluated using multiple parameters: Sodium adsorption ratio (SAR), percentage sodium (Na%), Kelly's ratio (KR), residual sodium carbonate (RSC), magnesium adsorption ratio (MAR), permeability index (PI) and the Corrosivity ratio (CR) (Acharya et al., 2018; Gugulothu et al., 2022; M'nassri et al., 2022). These parameters were respectively calculated using Eqs. (9) through (15), with ionic concentrations expressed in milliequivalents per litre (meq/L).

$$\text{SAR} = \frac{\text{Na}^+}{\sqrt{\frac{\text{Ca}^{2+}+\text{Mg}^{2+}}{2}}} \tag{9}$$

$$\%\text{Na} = \frac{\text{Na}^+ + \text{K}^+}{\text{Na}^+ + \text{K}^+ + \text{Ca}^{2+} + \text{Mg}^{2+}} x \frac{100}{1} \tag{10}$$

$$\text{KI} = \frac{\text{Na}^+}{\text{Ca}^{2+} + \text{Mg}^{2+}} \tag{11}$$

$$RSC = (HCO_3^- + CO_3^{2-}) - \left(Ca^{2+} + Mg^{2+}\right) \tag{12}$$

$$\text{MH} = \frac{\text{Mg}^{2+}}{\text{Ca}^{2+} + \text{Mg}^{2+}} \text{X} \frac{100}{1} \tag{13}$$

$$\text{PI} = \frac{\text{Na}^+ + \sqrt{\text{HCO}_3^-}}{\text{Na}^+ + \text{Ca}^{2+} + \text{Mg}^{2+}} \frac{100}{1} \tag{14}$$

$$\text{CR} = \frac{\left(\frac{\text{Cl}^-}{35}\right) + \left(\frac{\text{SO}_4^{2-}}{48}\right)}{\left(\frac{\text{CO}_3^{2-}+\text{HCO}_3^-}{50}\right)} \tag{15}$$

The study also used the irrigation water quality index (IWQI) method to determine the suitability of springwater for irrigation purposes. This index serves as an integrated measure of irrigation water quality and was evaluated using seven key parameters: EC, SAR, %Na^+, KI, RSC, MAR, PI (Sanad et al., 2024). Table 2 presents the parameters, the weight and relative weights of each parameter. Parameters crucial for water quality assessment, such as EC and SAR, received the highest rating of 5, while those with minimal significance, like RSC, were assigned just 1. The remaining parameters were rated between 2 and 4 based on their relative importance in evaluating water quality. Figure 2 represents the process flow diagram of the methodology used in this study.

Table 2 Irrigation parameters and their weights and relative weights

Parameter	Minimum	Maximum	Mean	Limit for irrigation use	Weight	Relative weight
EC	38.39	254.32	132.68	2250	5	0.2083
SAR	0.06	1.12	0.36	18	5	0.2083
%Na	4.77	61.42	26.11	60	3	0.125
KR	0.04	1.25	0.29	1	3	0.125
RSC	− 1.62	− 0.02	− 0.53	2.5	1	0.0417
MAR	17.24	96.79	60.81	50	3	0.125
PI	40.00	117.41	78.42	75	4	0.1667
			Total		24	1

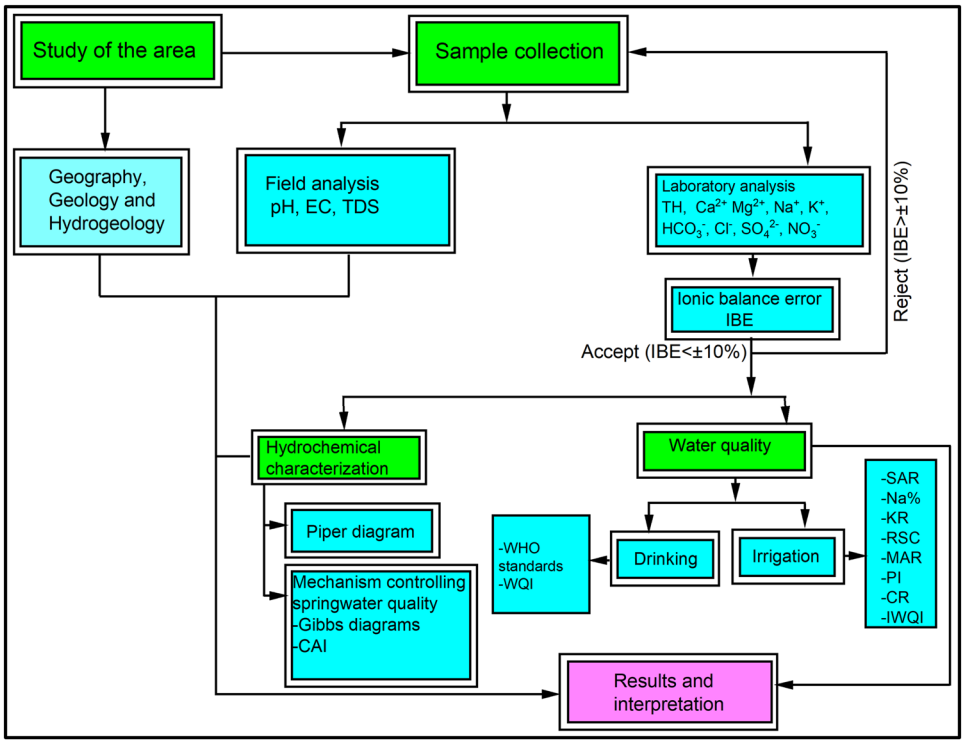

Fig. 2 Flow chart of research methodology

Results and discussion

Springwater chemistry and suitability for drinking purposes

The pH of the analysed spring samples ranged from 4.2 to 7.7, with an average value of 5.93 (Table 1). The pH levels in the study area were slightly acidic, with only 15 samples falling within the WHO guidelines. The range of values of pH in this study is consistent with those reported by Akaho et al. (2022) in Baham for groundwater and streams. The predominantly low pH observed in the sampled springs suggests the existence of weakly ionising acids, elevated CO_2 levels, and oligotrophic water conditions. Furthermore, the breakdown of accumulated organic materials like leaf litter in the dry season contributes to the low pH levels at these locations. Microbial oxidation of organic matter leads to the release of CO_2 and other byproducts like amino acids, which decrease pH (Kasa et al., 2022). The spatial distribution map of pH in the study area is found in Fig. 3a. Electrical Conductivity (EC), which indicates water's ability to conduct electricity, ranged from 38.39 to 254.32 µS/cm with an average of 132.68 µS/cm. All EC values were below their WHO standard limits of 1000 µS/cm. Total dissolved solids (TDS) varied from 22 to 151 mg/L, averaging 78.92 mg/L, with no sample falling out of the acceptable limit of 500 mg/L. It is noteworthy that consuming water with TDS values exceeding 500 mg/L can lead to gastrointestinal irritation (Adjovu et al., 2023). The total hardness (TH) concentration ranged from 11.30 to 111.24 mg/L and were all within their WHO acceptable limits (500 mg/L). Based on the TH values, the groundwater samples were classified into two categories: 37.84% of samples were classified as 'moderately hard', and the majority (62.16%) fell into the 'soft' category (McGowan, 2000). Hardness is mainly attributed to the presence of magnesium and calcium in the spring water. At higher concentrations, it can lead to the formation of kidney stones (Chanyalew et al., 2025). Figure 3b reveals that moderately hard water is mainly found in Baham.

Chemical analysis showed a clear hierarchical pattern of ion concentrations, with major cations following the order $Ca^{2+} > Mg^{2+} > Na^{+} > K^{+}$. Specifically, concentration ranges were: Na^{+} (0.81–13.40 mg/L), K^{+} (0.13–8.95 mg/L), Ca^{2+} (0.62–20.85 mg/L), and Mg^{2+} (0.74–15.41 mg/L). Na^{+} emerged as the dominant alkali metal, likely originating from halite and silicate mineral weathering during the movement of the spring water to the surface, with possible contributions from agricultural activities. Na^{+} showed higher concentrations than K^{+} due to its superior dissolution rate, while K^{+} presence was attributed to potassium-based fertilizers and decomposing organic matter. Ca^{2+} was the dominant alkaline earth metal, and Mg^{2+} was linked to dolomite and ferromagnesium mineral weathering. Notably, all cationic concentrations fell within WHO's permissible limits for both drinking and irrigation purposes.

Anionic concentrations followed the order $HCO_3^- > Cl^- > NO_3^- > SO_4^{2-}$. Cl^- concentrations in the study area varied from 6.13 to 44.95 mg/L, with a mean

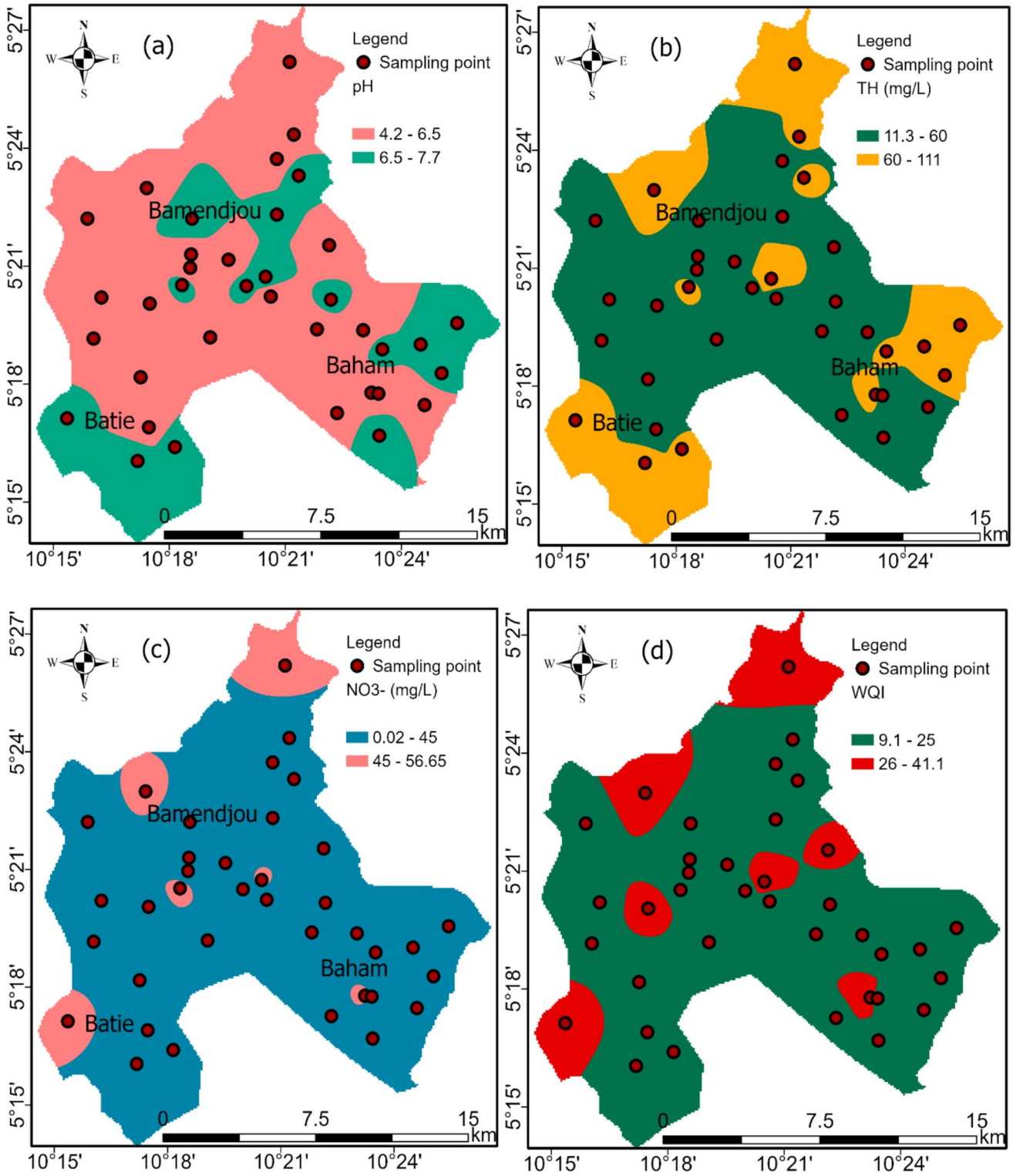

Fig. 3 Spatial distribution of springwater quality parameters and water quality index (WQI): **a** pH, **b** TH, **c** NO_3^- **d** WQI

value of 20.62 mg/L, which falls well below the maximum permissible limit of 250 mg/L. Variations in Cl^- levels can be attributed to both natural factors (including geological differences, rainfall patterns, dissolution of fluid inclusions, and Cl-bearing minerals) and anthropogenic sources (such as domestic effluent and the use of fertilizers) in the

study area (Nasher & Ahmed, 2021). HCO_3^- concentrations ranged from 3.54 to 79.15 mg/L, with an average value of 29.54 mg/L. In the study area, NO_3^- concentrations varied from 0.02 to 56.65 mg/L with an average value of 11.71 mg/L (Fig. 3c). NO_3^- contamination of water is a widespread global issue, caused by excessive nitrogen fertilizers, manure use, septic tank leakage, urbanisation, industrial and municipal sewage, atmospheric deposition, and landfill leachates (Adimalla & Qian, 2019). Due to the potential health risks associated with high NO_3^- levels, the WHO has established a standard limit of 45 mg/L to protect against health problems like methemoglobinemia and thyroid issues, especially in children. Figure 3c highlights that the highest concentrations of nitrate are observed in Bamendjou. The elevated levels of NO_3^- in the area can be attributed to unregulated human activities like inadequate management of sewage, agricultural compost practices, and disposal of animal waste (Li et al., 2019). SO_4^{2-} is another chemical parameter that affects drinking water quality, particularly its taste and odour characteristics. The analysed springwater samples showed SO_4^{2-} concentrations ranging from 0.25 to 29.22 mg/L, with a mean value of 4.71 mg/L. These levels are well below the threshold (250 mg/L) that would cause noticeable taste or laxative effects in unaccustomed consumers.

The WQI was used to evaluate overall springwater quality in the study area, using a classification system that defines five categories of drinking water quality: excellent (≤ 25), good (26–50), poor (51–75), extremely poor (76–100), and unsuitable for drinking (> 100) (Ameen, 2019). Water quality analysis yielded WQI values between 9.1 and 41.1 (mean 20.75), with 30 samples classified as excellent and the remainder as good, while none fell into poor or unsuitable categories (Fig. 3d). Despite these favorable ratings, the good water quality ratings were in line with pH levels below the WHO lower limit of 6.5 and NO_3^- concentrations surpassing the WHO standard of 45 mg/L, signaling growing concerns regarding pH and NO_3^- levels in the springwater quality. Therefore, elevated WQI values observed in certain samples could be linked to human activities such as intensive irrigation, extensive springwater extraction, and unregulated waste disposal, as well as natural processes like rock weathering and mineral dissolution in the area.

Hydrochemical facies

Hydrochemical facies analysis reveals the relationships and distribution patterns of major ions in water. Using the Piper diagram (Piper, 1944), springwater samples were classified into six distinct water types (from the diamond shaped fields) based on dominant ions: (1) Ca-HCO_3, (2) Na-Cl, (3) mixed Ca–Mg–Cl, (4) Ca–Cl, (5) mixed Ca-Na-HCO_3, and (6) Na-HCO_3 (Fig. 4). The results showed the mixed Ca–Mg–Cl as the predominant water type (70.3%), indicating that the concentration of alkali earths exceeded that of alkali. Additionally, Ca–Cl (13.5%), Ca-HCO_3 (10.81%), and Na-Cl types (5.4%) were observed. No Na-HCO_3 or Ca-Na-HCO_3 types were observed. The high presence of

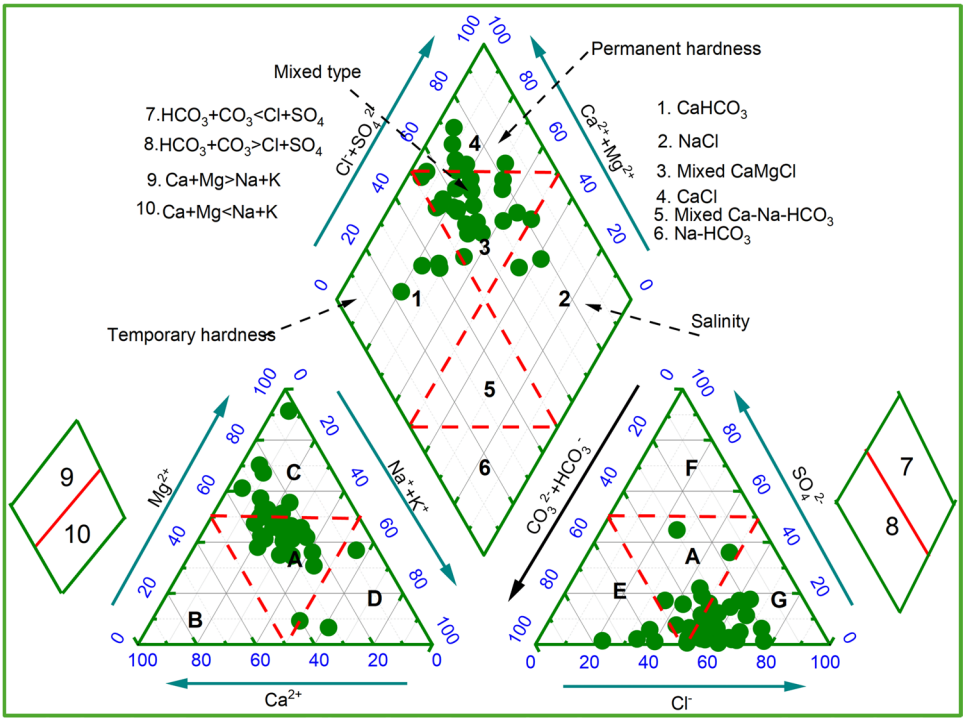

Fig. 4 Piper diagram showing springwater types in the study area

Ca–Mg–Cl indicates contamination from human activities like waste disposal, wastewater, and irrigation runoff (Nematollahi et al., 2016). Meanwhile, the Ca-HCO_3 and Ca–Cl facies types point to natural processes like mineral dissolution, rock-water interactions, and freshwater recharge. Na-Cl facies may be due to the dissolution of alkali feldspar minerals in the springwater (Selvakumar et al., 2017). Among the anions, strong acids were more prevalent than weak acidic anions. The lower right triangle shows that chloride ions were the dominant species (64.9%) in zone G, while bicarbonate (zone E) and no dominant species (zone A) were present in 10.8% and 24.3% of the samples, respectively. The high levels of chloride are likely due to precipitation processes, human activities, and rock salt dissolution. For cations (lower left triangle), most samples fell into the no dominance category (zone A) at 70.3%, followed by magnesium (zone C) at 24.32%, with sodium (zone D) appearing in 5.4% of the samples.

Mechanism controlling spring water quality

Gibbs diagrams analyze the chemical and lithological characteristics of an aquifer within groundwater, categorized into evaporates, precipitation, and rock zones. As per Gibbs (1970), these diagrams employ the correlations between TDS and $(Na^+ + K^+)$ / $(Na^+ + K^+ + Ca^{2+})$, as well as TDS and $(Cl^-)/(Cl^- + HCO_3^-)$ to ascertain the hydrogeochemical origins in springwater systems. Figure 5 reveals that 89.2% of the data points fall within the rock dominance zone, indicating that rock-water interactions are the primary factor influencing the springwater's chemical composition. However, some sampling points show a shift from rock dominance towards the rainfall zone, suggesting an increase in salt, chloride, and TDS concentrations in these locations caused by rainfall. The hydrogeochemical composition of the springwater, which is largely determined by rock-water interactions, has important implications for water quality and public health. In this study, the chemical signature of the springwater clearly reflects the dominance of rock-water interactions in the subsurface environment.

Chloroalkaline indices (CAI-I and CAI-II), introduced by Schoeller (1934), serve as a well-established method for evaluating ion exchanges occurring between spring water and the geological formations they traverse within the subsurface environment. These indices, computed through specific equations, help identify the nature of ion exchanges: positive CAI values signify the absence of base ion exchange, while negative values indicate the occurrence of such exchanges, involving Na^+ and K^+ in water trading places with Ca^{2+} and Mg^{2+} in the rocks. In this study, CAI-I values ranged from -0.540 to 0.909 (mean: 0.41), while CAI-II values ranged from -0.083 to 0.612 (mean: 0.22). It was found that most samples showed positive ratios (91.89% each for CAI-I and CAI-II), indicating no base-exchange reactions, whereas 8.1% displayed negative values, signifying active exchanges where Na^+ and K^+ in water are exchanged for Ca^{2+} and Mg^{2+} in the rock during their transit to the surface.

Irrigation water quality

Sodium Adsorption Ratio (SAR)

The Sodium Adsorption Ratio (SAR) measures the ratio of Na^+ to Ca^{2+} and Mg^{2+} in irrigation water. High SAR values create Na^+ hazards by reducing the availability of essential

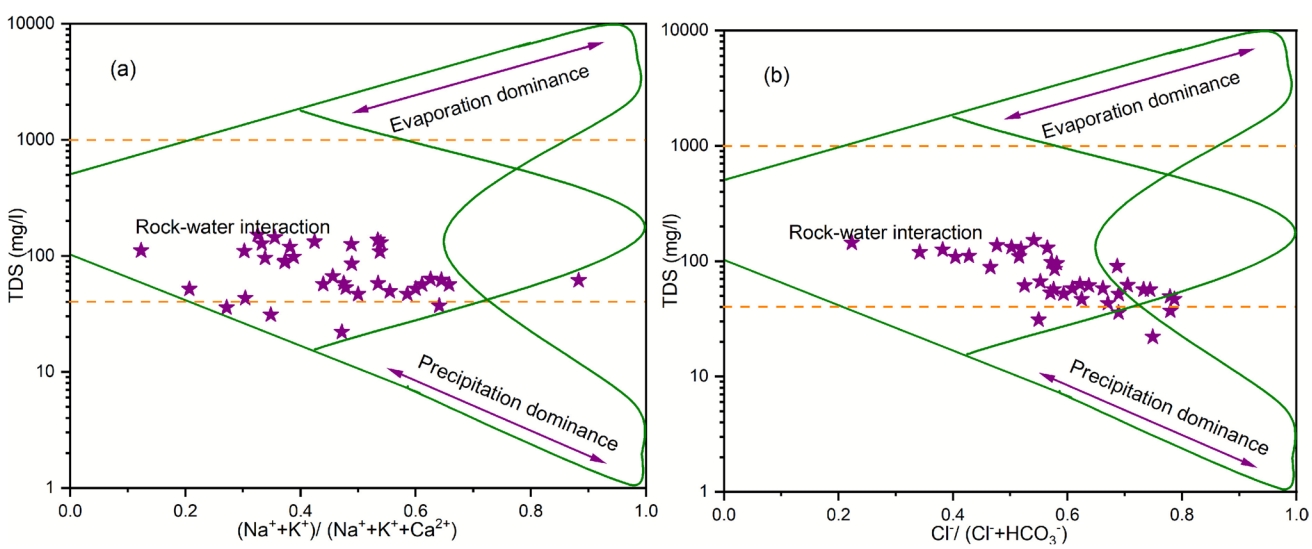

Fig. 5 Gibbs diagram showing mechanisms controlling springwater chemistry: **a** cations, **b** anions

Ca^{2+} and Mg^{2+} nutrients, which impairs crop growth. Irrigation water quality can be classified using SAR values according to Richards (1954): excellent (SAR < 10), good (SAR 10–18), doubtful (SAR 18–26), and unsuitable (SAR > 26) (Table 3). Consequently, springwater with SAR values above 26 is considered unsuitable for irrigation purposes. The analysed springwater samples showed SAR values between 0.06 and 1.12, indicating that all sampling points in the region are safe for agricultural use.

Percentage sodium (Na%)

The irrigation water quality was evaluated using sodium percentage (Na%), which is calculated from the concentrations of four dissolved cations (sodium, potassium, magnesium, and calcium) measured in milliequivalents per litre. High %Na values can negatively impact plant growth by reducing soil permeability, which restricts water and nutrient uptake by plant roots. Springwater classification for irrigation uses based on Na% is found in Table 3. The evaluated springwater samples revealed %Na values ranging from 4.77 to 61.42, with a mean of 26.11. These results indicate that 29.7%, 56.8%, 10.8% and 2.7% of the samples were in the excellent, good, permissible and doubtful ranges for irrigation purposes.

Table 3 Classification of irrigation water quality based on various indices (Chen et al., 2019; El-Rawy et al., 2019; Goswami et al., 2022)

Irrigation index	Class	Springwater status	Number of samples	% of samples
SAR	< 10	Excellent	37	100
	10–18	Good	–	–
	18–26	Doubtful	–	–
	> 26	Unsuitable	–	–
Na%	< 20	Excellent	12	32.43
	20–40	Good	20	54.05
	40–60	Permissible	4	10.8
	60–80	Doubtful	1	2.7
	> 80	Unsuitable	–	–
KR	< 1	Suitable	35	94.3
	> 1	Unsuitable	2	5.7
RSC	< 1.5	Good	37	100
	1.5–2.5	Doubtful	–	–
	> 2.5	Unsuitable	–	–
MAR	< 50	Suitable	3	8.1
	> 50	Unsuitable	34	91.9
CR	< 1	Suitable	4	10.8
	> 1	Unsuitable	33	89.2
IWQI	0–25	Excellent	–	–
	26–50	Good	31	86.5
	51–75	Poor	4	10.8
	76–100	Very poor	1	2.7
	> 100	Unsuitable	–	–

Kelly's ratio (KR)

Kelly's ratio (KR) measures the excess Na^{+} in water relative to Ca^{2+} and Mg^{2+} levels, which helps to determine springwater's irrigation suitability. According to Kelly, (1940), water with KR values below 1 is acceptable for irrigation, while values above 1 indicate excessive Na^{+} and make the water unsuitable. In this study, KR values ranged from 0.04 to 1.25 with a mean value of 0.29. It was found that about 94.6% of the samples (35 samples) had KR values within acceptable limits for irrigation use (Table 3).

Residual Sodium Carbonate (RSC)

RSC measures groundwater's suitability for irrigation by comparing levels of CO_3^{2-} and HCO_3^{-} with Ca^{2+} and Mg^{2+}. The RSC values fall into three categories (Ragunath, 1987): suitable (less than 1.25 meq/L), marginally suitable (1.25–2.50 meq/L), and unsuitable (above 2.50 meq/L). In the study area, all sampling sites were suitable for irrigation, with RSC values ranging from − 1.62 to − 0.02 meq/L and averaging − 0.53 meq/L.

Magnesium adsorption ratio (MAR)

Ca^{2+} and Mg^{2+} play crucial roles in soil structure, but their excessive concentrations can raise soil pH and reduce phosphorus availability. MAR in the studied samples ranges from 17.24 to 96.79, averaging 60.81. High Mg^{2+} levels specifically lead to increased soil alkalinity, which negatively affects crop productivity. Hence, MAR is used to measure magnesium's harmful effects and determine if water is suitable for irrigation purposes. According to established standards, water with MAR above 50% is considered harmful for irrigation, while values below 50% are suitable (Szabolcs & Darab, 1964). Hence, based on the results, only 8.1% of the study area has suitable water for irrigation, while 91.9% is unsuitable.

Corrosivity ratio (CR)

In this study, the corrosivity ratio (CR) was used to evaluate spring water safety for pipe transport, with values below 1 indicating non-corrosive water suitable for irrigation. Knowledge of CR serves as an economic indicator for irrigation supply system design, helping determine whether irrigation water can be safely transported using inexpensive PVC plastic pipes or more cost-effective metallic pipes. This information is especially critical for

long irrigation drainage systems, as low water levels in open drainage configurations increase contamination risks (Chauhdary et al., 2024). Results revealed that only 5.4% of the analysed spring water samples had CR values under 1 (Table 3), confirming their safety for irrigation purposes. Hence, based on the CR values of irrigation water, low-cost PVC piped delivery systems are recommended to minimize corrosion risk and optimize infrastructure investment.

Permeability index (PI)

Doneen's, 1964 classification assesses irrigation water quality using the Permeability Index (PI), which considers the long-term effects of Na^+, Ca^{2+}, Mg^{2+}, and HCO_3^- contents on soil permeability. The classification system defines three categories: Class 1 (excellent) and Class 2 (good), indicating water suitable for irrigation with 75% or more maximum permeability, while Class 3 (unsuitable) indicates water with 25% maximum permeability. In the study area, PI values ranged from 40 to 117.41, averaging 78.42. Analysis showed that 54.1% of samples fell in the "good" category, while 45.9% were classified as unsuitable (Fig. 6a).

Wilcox diagram

Wilcox's diagram evaluates irrigation water quality by plotting Na% against electrical conductivity (EC), categorizing water into five classes: excellent to good, good to permissible, permissible to doubtful, doubtful to unsuitable, and unsuitable (Fig. 6b). The analysed samples showed that all water samples fell into the excellent category. Hence, all studied samples are suitable for irrigation purposes.

Irrigation water quality index (IWQI)

The IWQI evaluates seven irrigation indices to determine their composite effects on irrigation water quality. Results revealed values ranging from 26.1 to 78.9, with an average value of 45.55 across all samples. The quality classification results showed that while the majority of samples (86.5%) achieved "good" status, a concerning portion fell into less favorable categories: 10.8% were classified as "poor" and 2.7% as "very poor." Notably, none of the samples achieved an "excellent" or " unsuitable" classification. Thus, based on

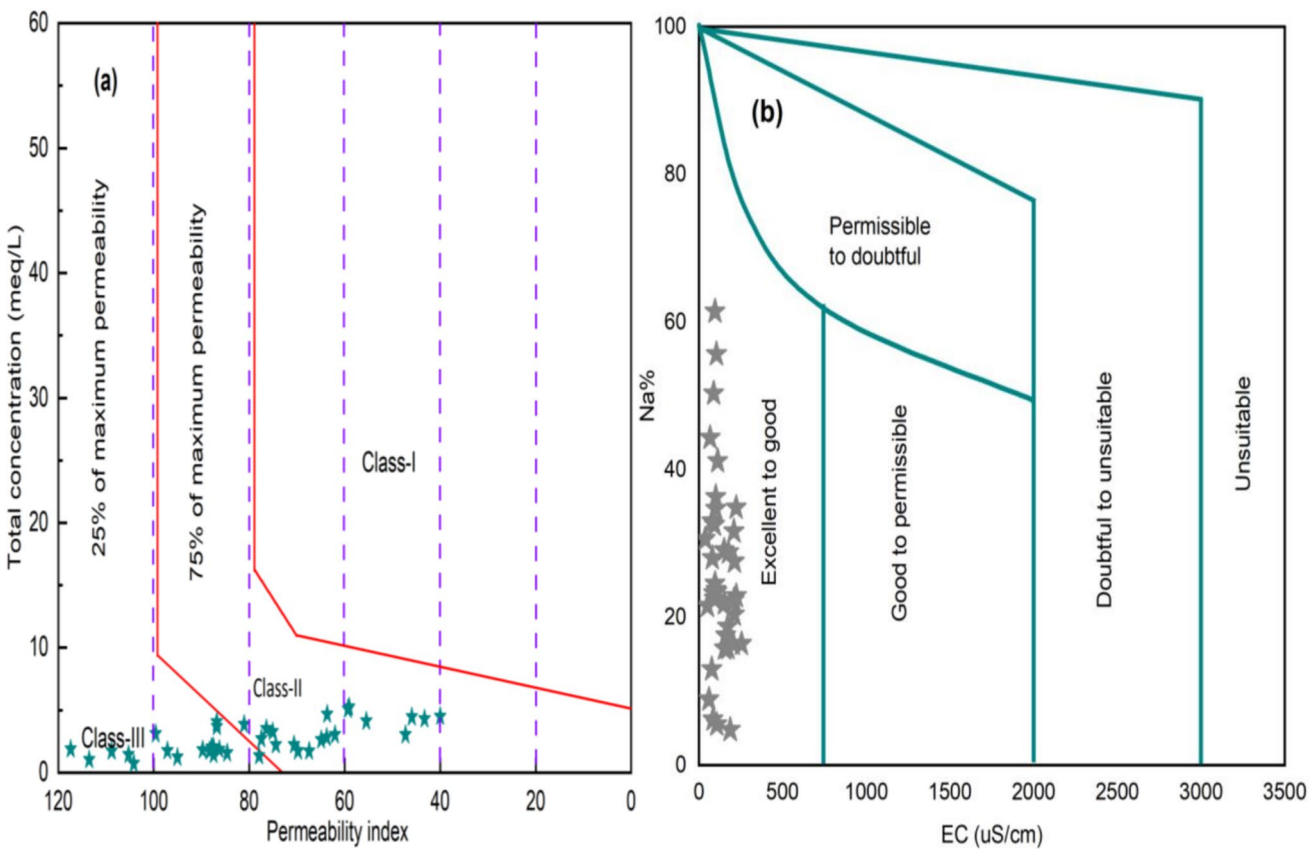

Fig. 6 **a** Doneen's permeability plot for irrigation water quality, **b** Wilcox diagram

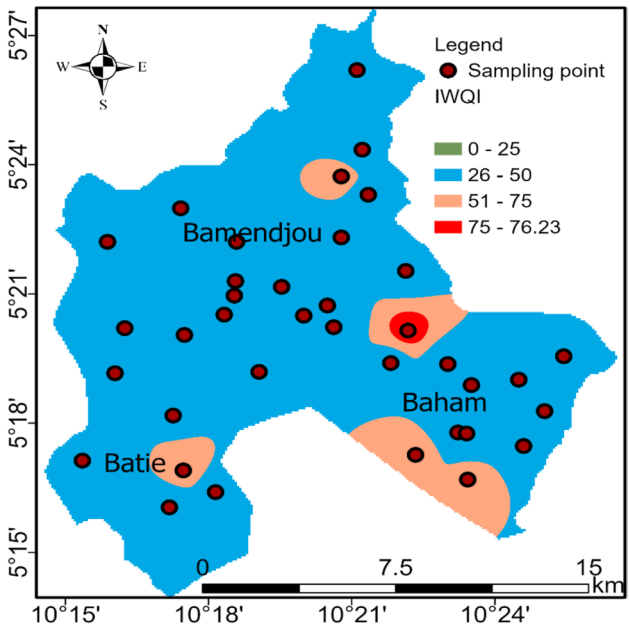

Fig. 7 Spatial distribution of IWQI for the study area

IWQI, the sampling points are generally good for irrigation quality. The spatial analysis of the IWQI (Fig. 7) reveals that regions with poor to very poor irrigation water are sparsely distributed across the study area, with three samples in Baham and one each in Bamendjou and Batie. These locations showed the impacts of human activities such as agricultural fertiliser use and wastewater discharge. Spring sample 5 had the highest IWQI, likely due to additional impacts of natural and geological factors.

Conclusion

Spring water constitutes a crucial water source in the Haut Plateau division, but growing population demands and climate change-induced rainfall reduction have stressed local water systems. Given that spring quality is fundamental to the community's health and development, the study focused on determining whether local springs meet the standards for safe drinking water and agricultural use. The results show that the spring water's pH levels range from 4.7 to 7.7 (acidic to basic), with low mineralization levels (EC from 38.39 to 254.32 µS/cm). All ionic concentrations in the water conformed to the WHO standards for drinking water quality, except for NO_3^-, where concentrations at some sampling points exceeded 45 mg/l. WQI assessment indicates that all sampled springwater sources meet excellent to good drinking water standards. However, analysis revealed two emerging challenges in spring water quality: acidic pH levels and rising nitrate concentrations due to human activities, though other parameters remained within acceptable limits. Further, the Piper plot demonstrates that in most spring samples, alkaline earth elements prevail over alkali metals, and strong acids exceed weak acids. Based on the findings, four water types were delineated: mixed Ca–Mg–Cl, Ca–Cl, Ca-HCO_3 and Na-Cl water types. The chemistry of the springwater in the study area is predominantly influenced by rock-water interaction and reverse ion exchange processes. Results based on irrigation indices indicate that MAR and PI were significant pollutants affecting irrigation water quality. Despite this, when considering the IWQI, it was found that 86.5% of the examined samples were in the good category for irrigation purposes. Additionally, the research demonstrated that only 5.4% of the springwater in the study area is considered corrosion-free when transported through metal pipes. Based on these findings, it is recommended that for low pH and high nitrate issues, limestone filters can be implemented to neutralise acidity and biological denitrification systems to convert nitrates to nitrogen gas. Furthermore, regular monitoring of irrigation water quality is essential for maintaining agricultural productivity, as these data enable farmers and policymakers to develop both effective irrigation guidelines and sustainable groundwater management plans that work together to ensure long-term farming success.

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Data availability Data is available upon reasonable request by the corresponding author.

Declarations

Conflict of interest The authors declare that they have no competing interests.

References

Abdessamed, D., Jodar-Abellan, A., Ghoneim, S. S. M., Almaliki, A., Hussein, E. E., & Pardo, M. Á. (2023). Groundwater quality assessment for sustainable human consumption in arid areas based on GIS and water quality index in the watershed of Ain Sefra (SW of Algeria). *Environmental Earth Sciences, 82*, 510. https://doi.org/10.1007/s12665-023-11183-9

Abduljabar, P., Hassan, N., & Karimi, H. (2020). Assessment of physicochemical parameters of spring water sources in Amediye District, Kurdistan Region of Iraq. *International Journal of Health and Life Sciences, 6*(1), Article e100324. https://doi.org/10.5812/ijhls.100324

Acharya, S., Sharma, S. K., & Khandegar, V. (2018). Assessment of groundwater quality by water quality indices for irrigation

and drinking in South West Delhi, India. *Data in Brief, 18*, 2019–2028. https://doi.org/10.1016/j.dib.2018.04.120

Adimalla, N., & Qian, H. (2019). Groundwater quality evaluation using water quality index (WQI) for drinking purposes and human health risk (HHR) assessment in an agricultural region of Nanganur, south India. *Ecotoxicology and Environmental Safety, 176*, 153–161. https://doi.org/10.1016/j.ecoenv.2019.03.066

Adjovu, G. E., Stephen, H., James, D., & Ahmad, S. (2023). Measurement of total dissolved solids and total suspended solids in water systems: A review of the issues, conventional, and remote sensing techniques. *Remote Sensing, 15*(14), 1–43. https://doi.org/10.3390/rs15143534

Ahada, C. P. S., & Suthar, S. (2018). Assessing groundwater hydrochemistry of Malwa Punjab. *India. Arabian Journal of Geosciences, 11*, 17. https://doi.org/10.1007/s12517-017-3355-8

Akaho, A. A., Tikeri, G. B., & David, A. O. (2022). Physicochemical analysis of potable water in Baham Community, Western Region of Cameroon. *Journal of Applied Sciences and Environmental Management, 26*(7), 1203–1209. https://doi.org/10.4314/jasem.v26i7.3

Al-Aizari, H. S., Aslaou, F., Mohsen, O., Al-Aizari, A. R., Al-Odayni, A. B., Abduh, N. A. Y., Al-Aizari, A. J. M., & Taleb, E. A. B. O. (2024). Assessment of groundwater quality for irrigation purpose using Irrigation Water Quality Index (Iwqi). *Journal of Environmental Engineering and Landscape Management, 32*(1), 1–11. https://doi.org/10.3846/jeelm.2024.20598

Alghamdi, A. G., Aly, A. A., Majrashi, M. A., & Ibrahim, H. M. (2023). Impact of climate change on hydrochemical properties and quality of groundwater for domestic and irrigation purposes in arid environment: A case study of Al-Baha region, Saudi Arabia. *Environmental Earth Sciences, 82*, Article 39. https://doi.org/10.1007/s12665-022-10731-z

Ameen, H. A. (2019). Spring water quality assessment using water quality index in villages of Barwari Bala, Duhok, Kurdistan Region, Iraq. *Applied Water Science, 9*, Article 176. https://doi.org/10.1007/s13201-019-1080-z

Aouiti, S., Azaza, F. H., Melki, F. E., Hamdi, M., Celico, F., & Zammouri, M. (2021). Groundwater quality assessment for different uses using various water quality indices in semi-arid region of central Tunisia. *Environmental Science and Pollution Research, 28*, 46669–46691. https://doi.org/10.1007/s11356-020-11149-5

APHA. (1995). *Standard Methods for the Examination of Water and Wastewater*. APHA-AWWA-WEF.

Brown, R. M., McClelland, N. I., Deininger, R. A., & Tozer, R. G. (1970). A water quality index-do we dare? *Water & Sewage WorKs, 117*, 339–343.

Chanyalew, B., Kasa, T., Wondimu, Z., & Beyene, D. (2025). Evaluation of water quality status and pollution source apportionment of Kulfo River adjacent to Arba Minch town. *International Journal of Energy and Water Resources, 9*, 1095–1112. https://doi.org/10.1007/s42108-024-00332-w

Chauhdary, J. N., Li, H., Jiang, Y., Pan, X., Hussain, Z., Javaid, M., & Rizwan, M. (2024). Advances in sprinkler irrigation: A review in the context of precision irrigation for crop production. *Agronomy, 14*, Article 47. https://doi.org/10.3390/agronomy14010047

Chen, J., Huang, Q., Lin, Y., Fang, Y., Qian, H., Liu, R., & Ma, H. (2019). Hydrogeochemical characteristics and quality assessment of groundwater in an irrigated region, Northwest China. *Water (Switzerland), 11*(1), 1–18. https://doi.org/10.3390/w11010096

Dehghani, M. H., Changani, F., & Baghal Asghari, F. (2022). Application of irrigation and drinking water quality indices to monitor ground water quality using geographic information system: A case study of the basins around Urmia Lake. *Iran. International Journal of Environmental Analytical Chemistry, 102*, 7380–7393. https://doi.org/10.1080/03067319.2020.1830982

Doneen, L. (1964). *Notes on water quality in agriculture. Published as a water science and engineering paper 4001, Department of Water Science and Engineering, University of California, Davis Eaton FM (1950) Significance of carbonate in irrigation water.*

Dumaru, B., Kayastha, S. P., & Pandey, V. P. (2021). Spring water assessment for quality and suitability for various uses: The case of Thuligaad watershed, western Nepal. *Environmental Earth Sciences, 80*, Article 586. https://doi.org/10.1007/s12665-021-09826-w

Egbueri, J. C. (2020). Heavy Metals pollution source identification and probabilistic health risk assessment of shallow groundwater in Onitsha, Nigeria. *Analytical Letters, 53*, 1620–1638. https://doi.org/10.1080/00032719.2020.1712606

El-Rawy, M., Ismail, E., & Abdalla, O. (2019). Assessment of groundwater quality using GIS, hydrogeochemistry, and factor statistical analysis in Qena governorate. *Egypt. Desalination and Water Treatment, 162*, 14–29. https://doi.org/10.5004/dwt.2019.24423

Fallatah, O., & Khattab, M. R. (2023). Evaluation of groundwater quality and suitability for irrigation purposes and human consumption in Saudi Arabia. *Water (Basel), 15*, 2352. https://doi.org/10.3390/w15132352

Fano, E., Brewster, M., & Thompson, T. (1986). Managing water quality in developing countries. *Natural Resources Forum, 10*(1), 77–87. https://doi.org/10.1111/j.1477-8947.1986.tb00781.x

Gibbs, R. J. (1970). Mechanisms controlling World Water chemistry. *Science, 170*(3962), 1088–1090.

Gopinath, G., Resmi, T. R., & Seralathan, P. (2013). Assessment of groundwater quality in Kavaratti Island in the Lakshadweep Archipelagos, India. *Chemistry and Ecology, 29*(4), 309–319. https://doi.org/10.1080/02757540.2012.760546

Goswami, R., Neog, N., & Thakur, R. (2022). Hydrogeochemical assessment of groundwater quality for drinking and irrigation in Biswanath and Sonitpur district of the Central Brahmaputra Plain, India. *Frontiers in Water, 4*, Article 889128. https://doi.org/10.3389/frwa.2022.889128

Gugulothu, S., Subbarao, N., Das, R., & Dhakate, R. (2022). Geochemical evaluation of groundwater and suitability of groundwater quality for irrigation purpose in an agricultural region of South India. *Applied Water Science, 12*, 142. https://doi.org/10.1007/s13201-022-01583-w

Howladar, M. F., Al Numanbakth, M. A., & Faruque, M. O. (2018). An application of Water Quality Index (WQI) and multivariate statistics to evaluate the water quality around Maddhapara Granite Mining Industrial Area, Dinajpur, Bangladesh. *Environmental Systems Research, 6*, Article 13. https://doi.org/10.1186/s40068-017-0090-9

Hwang, S.-O., Cho, I.-H., Kim, H.-K., Hwang, E.-A., Han, B.-H., & Kim, B.-H. (2024). Toward a brighter future: Enhanced sustainable methods for preventing algal blooms and improving water quality. *Hydrobiology, 3*(2), 100–118. https://doi.org/10.3390/hydrobiology3020008

Joelle, Y. E., Emile, T., Lucas, K., Jean-Paul, A., Mathieu, M. N., Brice, S. T. F., Robean, W. F., & Brice, T. K. (2021). Hydrogeochemistry of groundwater and surface water in Dschang town (West Cameroon): Alkali and alkaline-earth elements ascertain lithological and anthropogenic constraints. *Journal of Groundwater Science and Engineering, 9*(3), 212–224. https://doi.org/10.19637/j.cnki.2305-7068.2021.03.004

Kasa, T., Bassa, A. L., Negatu, G. T., Sahile, Z. A., & Reddythota, D. (2022). Investigation of Wabe River water fitness for agricultural and industrial purposes. *Heliyon, 8*, Article e11865. https://doi.org/10.1016/j.heliyon.2022.e11865

Kelly, W. (1940). Permissible composition and concentration of irrigated waters. *Proceeding of the ASCF*, 607.

Kumar, S., Rajesh, V., & Khan, N. (2022). Evaluation of groundwater quality in Ramanathapuram district, using water quality index

(WQI). *Modeling Earth Systems and Environment, 8*, 35–45. https://doi.org/10.1007/s40808-020-01025-z

Lanjwani, M. F., Khuhawar, M. Y., & Jahangir Khuhawar, T. M. (2022). Assessment of groundwater quality for drinking and irrigation uses in taluka Ratodero, district Larkana, Sindh, Pakistan. *International Journal of Environmental Analytical Chemistry, 102*, 4134–4157. https://doi.org/10.1080/03067319.2020.1780222

Li, C., Gao, X., Liu, Y., & Wang, Y. (2019). Impact of anthropogenic activities on the enrichment of fluoride and salinity in groundwater in the Yuncheng Basin constrained by Cl/Br ratio, δ18O, δ2H, δ13C and δ7Li isotopes. *Journal of Hydrology, 579*, Article 124211. https://doi.org/10.1016/j.jhydrol.2019.124211

M'nassri, S., El Amri, A., Nasri, N., & Majdoub, R. (2022). Estimation of irrigation water quality index in a semi-arid environment using data-driven approach. *Water Supply, 22*(5), 5161–5175. https://doi.org/10.2166/ws.2022.157

Makhlouf, A., Rawy, M. E., Kanae, S., Sharaan, M., Nada, A., & Ibrahim, M. G. (2025). Streamlining the monitoring and assessment of irrigation groundwater quality using machine learning techniques. *Environmental Earth Sciences, 84*, Article 148. https://doi.org/10.1007/s12665-025-12132-4

Marghade, D., Malpe, D. B., & Subba Rao, N. (2021). Applications of geochemical and multivariate statistical approaches for the evaluation of groundwater quality and human health risks in a semi-arid region of eastern Maharashtra, India. *Environmental Geochemistry and Health, 43*, 683–703. https://doi.org/10.1007/s10653-019-00478-1

Mba, F. F., Temgoua, E., Kengne, P. D., & Natheu Kamhoua, S. (2019). Vulnerability of groundwater to pollution in the city of Dschang, West Cameroon. *International Journal of Biological and Chemical Sciences, 13*(5), 39–56. https://doi.org/10.4314/ijbcs.v13i5.3S

Mbinkong, R. S., Kengni, S. H. P., Ndoh, N. E., Gaetan, T. D. N., Pokam, B. P. G., & Tabod, C. T. (2024a). Evaluation of groundwater potential, aquifer parameters and vulnerability using geoelectrical method: A case study of parts of the Sanaga Maritime Division, Douala, Cameroon. *Modeling Earth Systems and Environment*. https://doi.org/10.1007/s40808-023-01932-x

Mbinkong, R. S., Ndoh, N. E., Tabod, C. T., Mufur, A. M., Kengni, S. H. P., Gaetan, T. D. N., & Pokam, B. P. G. (2024b). Assessing groundwater suitability and nitrate health risk in Edea, Cameroon: Implications for drinking and irrigation purposes. *Modeling Earth Systems and Environment*. https://doi.org/10.1007/s40808-024-02116-x

McGowan, W. (2000). *Water processing: residential, commercial, light-industrial* (3rd edn.). Water Quality Association.

Mfonka, Z., Kpoumié, A., Ngouh, A. N., Mouncherou, O. F., Nsangou, D., Rakotondrabe, F., Takounjou, A. F., Zammouri, M., Ngoupayou, J. R. N., & Ndjigui, P.-D. (2021). Water quality assessment in the Bamoun Plateau, Western-Cameroon: Hydrogeochemical modelling and multivariate statistical analysis approach. *Journal of Water Resource and Protection, 13*, 112–138. https://doi.org/10.4236/jwarp.2021.132007

Mishra, R. K., Mentha, S. S., Misra, Y., & Dwivedi, N. (2023). Emerging pollutants of severe environmental concern in water and wastewater: A comprehensive review on current developments and future research. *Water-Energy Nexus, 6*, 74–95. https://doi.org/10.1016/j.wen.2023.08.002

Mohammadi, A. A., Morovati, M., Najafi Saleh, H., Ghale Askari, S., Ghaderpoori, M., Shams, M., Kalteh, S., & Sadeghi, S. (2022). Groundwater quality evaluation for drinking and industrial purposes. A case study in Northeastern Iran. *International Journal of Environmental Analytical Chemistry, 102*, 6094–6104. https://doi.org/10.1080/03067319.2020.1807533

Mufur, M. A., Awah, M. T., Tamfuh, P. A., Nyangang, A. J., Njei, A. G., Kabeyene, K., & Beyela, V. (2022). Physico-chemical and bacteriological characterization of groundwater for domestic and irrigation purposes in Santchou (Cameroon Western Highlands). *American Journal of Environmental Protection, 11*(3), 62–73. https://doi.org/10.11648/j.ajep.20221103.12

Nematollahi, M. J., Ebrahimi, P., Razmara, M., & Ghasemi, A. (2016). Hydrogeochemical investigations and groundwater quality assessment of Torbat-Zaveh plain, Khorasan Razavi. *Iran. Environmental Monitoring and Assessment, 188*, 2. https://doi.org/10.1007/s10661-015-4968-6

Nong, X., Shao, D., Zhong, H., & Liang, J. (2020). Evaluation of water quality in the South-to-North Water Diversion Project of China using the water quality index (WQI) method. *Water Research, 178*, Article 115781. https://doi.org/10.1016/j.watres.2020.115781

Nono, A., Likeng, J., Wabo, H., Tabue Youmbi, G., & Biaya, S. (2009). Influence de la nature lithologique et des structures géologiques sur la qualité et la dynamique des eaux souterraines dans les hauts plateaux de l'Ouest-Cameroun. *International Journal of Biological and Chemical Sciences, 3*(2), 218–239. https://doi.org/10.4314/ijbcs.v3i2.44516

Nzouemou, F., Dongmo, N. G. T., Tabod, T. C., Ndoh, E. N., Mbinkong, R. S., & Kengni, S. H. P. (2025). Geoelectrical assessment of groundwater potential and aquifer characteristics in the west region of Cameroon. *Modeling Earth Systems and Environment, 11*, 7. https://doi.org/10.1007/s40808-024-02240-8

Piper, A. M. (1944). A graphic procedure in the geochemical interpretation of water analyses. *Atransactions, American Geophysical Union, 25*(6), 914–928.

Ragunath, H. M. (1987). *Groundwater* (2nd ed.). Wiley Eastern Ltd.

Rawat, K. S., Kumar, S., Sandeep, S., & Gautam, K. (2018). Assessment of groundwater quality for irrigation use : A peninsular case study. *Applied Water Science, 8*(8), 1–24. https://doi.org/10.1007/s13201-018-0866-8

Refat Nasher, N. M., & Humayan Ahmed, M. (2021). Groundwater geochemistry and hydrogeochemical processes in the Lower Ganges-Brahmaputra-Meghna River Basin areas, Bangladesh. *Journal of Asian Earth Sciences: X, 6*, Article 100062. https://doi.org/10.1016/j.jaesx.2021.100062

Richards, L. A. (1954). *Diagnosis and Improvement of Saline and Alkaline Soils*. Agricultural, Handbook 60. US Department of Agriculture, Washington, DC. https://doi.org/10.2136/sssaj1954.03615995001800030032x

Sanad, H., Mouhir, L., Zouahri, A., Moussadek, R., El Azhari, H., Yachou, H., Ghanimi, A., Oueld Lhaj, M., & Dakak, H. (2024). Assessment of groundwater quality using the Pollution Index of Groundwater (PIG), Nitrate Pollution Index (NPI), Water Quality Index (WQI), Multivariate Statistical Analysis (MSA), and GIS approaches: A case study of the Mnasra Region, Gharb Plain, Morocc. *Water, 16*, Article 1263. https://doi.org/10.3390/w16091263

Sanou, S. M., Temgoua, E., Guetiya, W. R., Arienzo, A., Losito, F., Fokam, J., Onohiol, J. F., Djeunang, B., Zambou, N. F., Russo, G., Antonini, G., Panà, A., & Colizzi, V. (2015). Water supply, sanitation and health risks in Douala 5 municipality, Cameroon. *Igiene e Sanità Pubblica, 71*(1), 21–37.

Schoeller, H. (1934). Les échanges de bases dans les eaux souterraines; trois exemples en Tunisie. *Bulletin De La Société Géologique De France, 4*, 389–420.

Selvakumar, S., Chandrasekar, N., & Kumar, G. (2017). Hydrogeochemical characteristics and groundwater contamination in the rapid urban development areas of Coimbatore, India. *Water Resources and Industry, 17*, 26–33. https://doi.org/10.1016/j.wri.2017.02.002

Sokeng, V.-C.J., Kouamé, F. K., N'da, H. D., Tankoano, B., You, L. A., & Ngatcha, B. N. (2016). Cartographie de l'occupation de sol des Hauts Plateaux de l'Ouest Cameroun par réseaux de neurones appliqués à une image LANDSAT 8 OLI [Mapping land cover in

Western Cameroon Highlands using neural networks applied to a Landsat 8 OLI image]. *International Journal of Innovation and Scientific Research, 23*(2), 443–454.

Sultana, J., Jolly, Y. N., Muktadir, M. G., Akter, S., Hossain, S., Mamun, K. M., & Fatema, J. (2025). Assessment of groundwater quality of a Tannery Industrial Estate of Dhaka, Bangladesh. *International Journal of Energy and Water Resources*. https://doi.org/10.1007/s42108-025-00365-9

Szabolcs, I., & Darab, C. (1964). The influence of irrigation water of high sodium carbonate content of soils. *Proceedings of 8th International Congress of ISSS, Trans, II*, pp. 803–812.

Taloor, A. K., Pir, R. A., Adimalla, N., Ali, S., Manhas, D. S., Roy, S., & Singh, A. K. (2020). Spring water quality and discharge assessment in the Basantar watershed of Jammu Himalaya using geographic information system (GIS) and water quality Index(WQI). *Groundwater for Sustainable Development, 10*, Article 100364. https://doi.org/10.1016/j.gsd.2020.100364

Temgoua, E. (2011). Chemical and bacteriological analysis of drinking water from alternative sources in the Dschang Municipality, Cameroon. *Journal of Environmental Protection, 02*, 620–628. https://doi.org/10.4236/jep.2011.25071

Temgoua, E., Ngnikam, E., & Ndongson, B. (2009). Drinking water quality: Stakes of control and sanitation in the town of Dschang - Cameroon. *International Journal of Biological and Chemical Sciences, 3*(3), 441–447. https://doi.org/10.4314/ijbcs.v3i3.45346

Ustaoğlu, F., Tepe, Y., & Taş, B. (2020). Assessment of stream quality and health risk in a subtropical Turkey river system: A combined approach using statistical analysis and water quality index. *Ecological Indicators, 113*, Article 105815. https://doi.org/10.1016/j.ecolind.2019.105815

von Fumetti, S., Bieri-Wigger, F., & Nagel, P. (2017). Temperature variability and its influence on macroinvertebrate assemblages of alpine springs. *Ecohydrology*. https://doi.org/10.1002/eco.1878

WHO. (2011). *Guidelines for drinking-water quality (4th ed., Vol. 564)*. Geneva: World Health Organization.

Xu, P., Feng, W., Qian, H., & Zhang, Q. (2019). Hydrogeochemical characterization and irrigation quality assessment of shallow groundwater in the central-Western Guanzhong basin, China. *International Journal of Environmental Research and Public Health, 16*, 1492. https://doi.org/10.3390/ijerph16091492

Zemunac, R., Savic, R., Blagojevic, B., Benka, P., Bezdan, A., & Salvai, A. (2021). Assessment of surface and groundwater quality for irrigation purposes in the Danube-Tisa-Danube hydrosystem area (Serbia). *Environmental Monitoring and Assessment, 193*, 193–519. https://doi.org/10.1007/s10661-021-09294-6

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